

BENCHMARK: NON-DETECTION

Results

Mid-Transit Time (Tmid)	2460289.721 +/- 0.0062 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.06 +/- 0.059
Transit depth (Rp/Rs)^2	0.36 +/- 0.71 [%]
Orbital Inclination (inc)	87.01 +/- 1.89
Airmass coefficient 1 (a1)	1.78 +/- 1.0
Airmass coefficient 2 (a2)	-0.41 +/- 0.4
Scatter in the residuals of the lightcurve fit is	50.76 %
Best Comparison Star	None
Optimal Aperture	0.87
Optimal Annulus	4.75
Transit Duration (day)	0.063 +/- 0.027

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.06 ± 0.059 vs 0.1592 (deviation 1.67σ)
Duration fit vs published	0.063 d vs 0.0946 d (deviation 1.16σ)
Mid-time O–C	15.5 min
Depth SNR	1.0
Residual scatter	50.76 %

Light curve

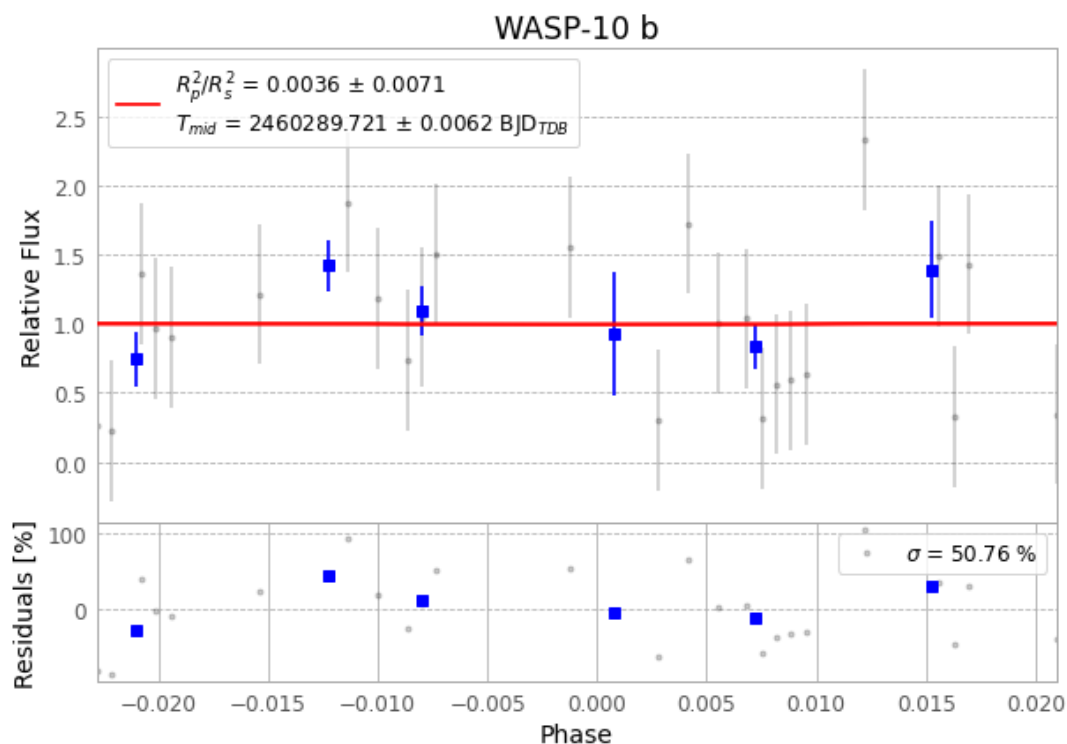


Figure 1 — FinalLightCurve_WASP-10 b_2023-12-10.png (SHA-256 c588962026b9dd91...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [153, 256], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 7bc0546a1a2289cf...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit 62f7813

Data provenance

76 FITS frames (50 MB), WASP-10231211031817.FITS → WASP-10231211070915.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-10-231211.log (7887663c1387...), FinalLightCurve_WASP-10 b_2023-12-10.png (c588962026b9...), FinalLightCurve_WASP-10 b_2023-12-10.csv (c122866fcfb6...)

Compute resources

Wall time 120.3 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-27 20:36Z.